

Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ without drift term

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Abstract

This paper deals with the long term behavior of the solution to the nonlinear stochastic heat equation $\partial u / \partial t - \frac{1}{2} \Delta u = b(u) \dot{W}$ for a Lipschitz continuous diffusion coefficient b which is not necessarily bounded such as $b(u) = u$. Using the theory of the stochastic integral laid out by John Walsh, we provide conditions which will guarantee the existence of an invariant measure for a broad range of initial conditions which includes L^2_ρ as well as the Dirac delta distribution δ_0 .

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1 Introduction

We consider the following *stochastic heat equation* (SHE):

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta u(t, x) = b(x, u(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0, \\ u(0, \cdot) = \mu(\cdot). \end{cases} \quad (1.1) \quad \boxed{\text{E:SHE}}$$

The noise, $\dot{W}(t, s)$, is a centered Gaussian noise that is white in time and homogeneously colored in space defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with natural filtration $(\mathcal{F}_t)$:

$$\mathcal{F}_t = \sigma(W_s(A) : 0 \leq s \leq t, A \in \mathcal{B}_b(\mathbb{R}^d)) \vee \mathcal{N},$$

where $\mathcal{B}_b(\mathbb{R}^d)$ denotes the bounded Borel sets on $\mathbb{R}^d$ and $\mathcal{N}$ are the $\mathbb{P}$ -null sets of Ω . Its covariance structure, J , is defined on as follows:

$$J(\psi, \phi) := \mathbb{E} [\dot{W}(\psi) \dot{W}(\phi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\psi(s, \cdot) * \tilde{\phi}(s, \cdot))(x), \quad (1.2) \quad \boxed{\text{E:Cor}}$$

for all continuous and rapidly decreasing functions ψ and ϕ , where $\tilde{\phi}(x) := \phi(-x)$, “ $*$ ” refers to the convolution in spatial variable, and Γ is a nonnegative and nonnegative definite tempered measure on $\mathbb{R}^d$ that is commonly referred to as the *correlation measure*. The Fourier transform of Γ (in the generalized sense) is also a nonnegative and nonnegative definite tempered measure,

which is called *spectral measure* and is denoted by $\widehat{f}(d\xi)$ ¹. Moreover, in the case where Γ has density f , namely, $\Gamma(dx) = f(x)dx$, then $\widehat{f}(d\xi) = \widehat{f}(\xi)d\xi$. The initial condition μ is a deterministic, locally finite, regular, signed Borel measure that satisfies some integrability condition at the infinity (see (3.4) below).

Any jointly measurable $L^2(\Omega)$ -continuous $\mathcal{F}_t$ -adapted processes, $u = (u(t, x) : t \geq 0, x \in \mathbb{R}^d)$, that solves almost surely the following stochastic integral equation is referred to as the *mild solution* to (1.1):

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) + \int_0^t \int_{\mathbb{R}^d} b(y, u(s, y)) G(t - s, x - y) W(ds, dy), \quad (1.3) \quad \boxed{\text{E:gmsol}}$$

where $G(t, x) = (2\pi t)^{-d/2} \exp\left(-(2t)^{-1}|x|^2\right)$ is the heat kernel and the stochastic integral is the *Walsh integral* (see [Wal86; Dal+09]).

Regarding the function $b(t, x)$, we assume that for some constants $L_b > 0$ and $L_0 \geq 0$, it holds that

$$|b(x, u) - b(x, v)| < L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0 \quad \text{for all } u, v \in \mathbb{R}, x \in \mathbb{R}^d. \quad (1.4) \quad \boxed{\text{E:lipcon}}$$

In particular, our assumption allows the linear case $b(x, u) = \lambda u$, which is usually referred to the *parabolic Anderson model* [CM94].

The aim of this paper is to investigate the conditions required to guarantee the existence of invariant measures for the solution to (1.1), which is a crucial step towards the study of the ergodicity of the system. Let ρ be an admissible weight function (see Definition 3.1) and we will work under the Hilbert space $H = L^2_\rho(\mathbb{R}^d)$. Let $\mathcal{B}(H)$ be the space of all bounded Borel functions on H . Following [DGZ92], a probability measure η on the Borel σ -field $\mathcal{B}(H)$ is said to be invariant for (1.1) if

$$\eta(A) = \int_H \mathcal{L}(u(t, \cdot; \mu))(A) \eta(d\zeta), \quad \text{for all } t > 0 \text{ and } A \in \mathcal{B}(H), \quad (1.5) \quad \boxed{\text{E:InvMeas}}$$

where we use the notation that $u(t, x; \varphi)$ with a third argument to emphasize the dependence of the solution to (1.1) on the initial condition $u(0, \cdot) = \varphi$. Note that in (1.5) we require t to be strictly bigger than zero, unlike Eq. (3) of [DGZ92] where one imposes $t \geq 0$.

Due the *Krylov-Bogoliubov theorem* (see [DZ96, Theorem 3.1.1]), we see that that existence of the measure can depend on the initial condition and under suitable conditions, for example the assumptions of Theorem 1.3 below, the following measure, η , is invariant for (1.1):

$$\eta(A) = \lim_{n \rightarrow \infty} \frac{1}{T_n} \int_{t_0}^{T_n + t_0} \mathcal{L}(u(t, \cdot, \varphi))(A) dt, \quad \text{for any } t_0 > 0. \quad (1.6) \quad \boxed{\text{E:InvMeasForm}}$$

In the above, $\{T_n\}$ is an appropriately chosen sequence of positive numbers increasing to infinity.

We note that proving uniqueness of the invariant measure is a much harder question. An open question is if φ_1 and φ_2 are two such initial conditions that are suitable for (1.6) and if η_1 and η_2 are the two invariant measures that arise from φ_1 and φ_2 respectively then does $\eta_1 = \eta_2$? Another question one can ask in the absence of uniqueness is what initial data are applicable to (1.6) and can we give explicit examples? We address the second question in Examples 2.5 and ?? and also in the discussion below.

Existence in unbounded domains has been studied in few papers such as [AM03; MSY20; MSY16; TZ98]. Since the spatial domain is unbounded, one needs to introduce a weight function

¹See Remark 2.8 for the convention of Fourier transform that is used in this paper.

ρ and the corresponding weighted space $L^2_\rho(\mathbb{R}^d)$ as usual; see Section 3 below for more details. Following the standard procedure as laid out by Da Prato and Zabczyk [DGZ92], the main effort is to show that:

$$\sup_{t>0} \mathbb{E} \left(\|u(t, \cdot)\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty. \quad (1.7) \quad \boxed{\text{E:BddMnt}}$$

However, this solution is usually *intermittent*, namely, the probability moments to (1.1) have a certain exponential growth in t . In [AM03; MSY20; MSY16], a drift term $f(x, u)$ is included in (1.1) in a crucial way to help to cancel the otherwise exponential growing moments. The absence of such a drift term in this paper makes the problem more challenging. In order to have moments bounded in time, as required in (1.7), one has to first identify the exact conditions, under which the moments to (1.1) do not possess exponential growth. This question has been answered in [CK19, Theorem 1.3 and Lemma 2.5], where necessary and sufficient conditions are given. More precisely, for bounded second moments one has to have the spatial dimension $d \geq 3$, and in addition, the spectral measure, $\hat{f}$, and Lipschitz constant, L_b , (the Lyapunov exponent of b) in (1.4) must satisfy the following:

$$\Upsilon(0) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{|\xi|^2} < \infty \quad \text{and} \quad 64L_b^2 < \frac{1}{4\Upsilon(0)}, \quad (1.8) \quad \boxed{\text{E:U0finite}}$$

where the first condition is a strengthened version of *Dalang's condition*:

$$\Upsilon(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty, \quad \text{for some (and hence) all } \beta > 0. \quad (1.9) \quad \boxed{\text{E:Dalang}}$$

Note that conditions in (1.8) guarantee the existence of the following non-empty open interval:

$$(128L_b^2\Upsilon(0), 1/2) \neq \emptyset. \quad (1.10) \quad \boxed{\text{E:Interval}}$$

In the setting of equation (1.1), Tessitore and Zabczyk [TZ98] prove existence of an invariant measure for (1.1) in $L^2_\rho(\mathbb{R}^d)$ under the assumption that there exists a $\varphi \in L^2_\rho(\mathbb{R}^d) \cap L^2_{\tilde{\rho}}(\mathbb{R}^d)$ where $\rho(\tilde{\rho})^{-1} \in L^1(\mathbb{R}^d)$ and the solution starting from φ is bounded in probability in $L^2_{\tilde{\rho}}(\mathbb{R}^d)$ and also that the spectral density $\hat{f}$ satisfies

$$\hat{f} \in L^p(\mathbb{R}^d) \quad \text{where} \quad \frac{d-2}{d} < \frac{1}{p}; \quad (1.11) \quad \boxed{\text{E:CorTZ?}}$$

see Hypothesis 2.1 (*ibid.*). However as was illustrated in [TZ98, Theorem 3.3], in order to apply this theorem (or to have moments uniformly bounded in time), the following additional assumptions were imposed:

$$d \geq 3 \quad \text{and} \quad L_b^{-2} > \frac{\Gamma(d/2 - 1)2^{d/2-2}}{(2\pi)^{2d}} \int_{\mathbb{R}^d} \left(\left| \mathcal{F} \left(\sqrt{\hat{f}} \right) \right| * \left| \mathcal{F} \left(\sqrt{\hat{f}} \right) \right| \right) (\zeta) |\zeta|^{2-d} d\zeta, \quad (1.12) \quad \boxed{\text{E:TZspec}}$$

where Γ refers to the Gamma function; See Remark 2.8. With these assumptions, they were able to prove that (1.1) starting from the constant 1 satisfies (1.7) and thus is bounded in probability, verifying the existence of an invariant measure. Moreover, the measure takes the form of (1.6) above with $\varphi = 1$.

The additional assumptions given by Tessitore and Zabczyk parallel our assumptions (1.8) and (??). However, it is not at all trivial to check which spectral densities, $\hat{f}(d\xi)$, satisfy (1.12) and no such examples were given in [TZ98]. For example, it may not be obvious from (1.12) that one can not take the spectral density of the noise to be the Riesz kernel, $\hat{f}(\xi) = |\xi|^{d-\alpha}$, for

any $0 < \alpha < d$ but this can be easily seen from our condition (1.8). The condition that $d \geq 3$ forces us to only consider a colored noise as in (1.2) and so the loss of the ability to apply the Riesz kernel is unfortunate. However, as we will see in Example 2.5 below, we are able to apply the Bessel kernel, $B_s(x)$, with $s < d$, which has exponential decaying tails and power growth at the origin. So in absence of the Riesz kernel, we still may apply our results to a kernel, namely the Bessel kernel, that has origin blow up.

Moreover by applying a restart argument (see Remark ?? and Example 2.5 below), our method allows us to verify (1.7), and in return (1.6), for more general initial conditions $\mu(\cdot)$ which include:

$$\text{Dirac delta measure } \delta_0(dx) \quad \text{and} \quad \varphi(x)dx \text{ with } \varphi \in C^\infty(\mathbb{R}^d).$$

Thus, we can expand our choice of initial data to include singular distributions, which are outside the space of $L^2_\rho(\mathbb{R}^d)$. We will expand on this in the following remark.

StartNotation)?

Notation 1.1. Denote $\|\cdot\|_\rho := \|\cdot\|_{L^2_\rho(\mathbb{R}^d)}$ and $\|X\|_p := (\mathbb{E}(|X|^p))^{1/p}$. We use $J_0(t, x)$ to denote the solution to the homogeneous equation and $J_0(t, x; \mu)$ to emphasize the dependency on the initial condition:

$$J_0(t, x; \mu) := (G(t, \cdot) * \mu)(x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy). \quad (1.13) \text{ ?E:Ju?}$$

For any signed Borel measure μ on $\mathbb{R}^d$, let $\mu = \mu_+ - \mu_-$ be its *Hahn decomposition* and set

$$|\mu| = \mu_+ + \mu_- \quad \text{and} \quad \mu^* := 1 + |\mu|.$$

We first present a preliminary result which is used to prove Theorem 1.3 below.

(T:Lrho)

Theorem 1.2. Let $u_0(\cdot)$ be an $\mathcal{F}_0$ -measurable random variable and $u(t, x; u_0)$ be the solution to (1.1) starting from u_0 . Assume that

(i) $\rho : \mathbb{R}^d \rightarrow \mathbb{R}_+$ is an nonnegative $L^1(\mathbb{R}^d)$ function;

(ii) the initial condition u_0 satisfies

$$\mathcal{G}_\rho(t; u_0) := \int_{\mathbb{R}^d} \mathbb{E} (J_0^2(t, x; u_0^*)) \rho(x) dx < \infty \quad \text{for all } t > 0; \quad (1.14) \text{ [E:G]}$$

(iii) the correlation measure Γ and the Lipschitz constant L_b of b satisfy conditions in (1.8).

Then for all $t > 0$, $u(t, \cdot; u_0) \in L^2_\rho(\mathbb{R}^d)$ a.s. Moreover, for some constant $C > 0$ which does not depend on t , it holds that

$$\mathbb{E} \left(\|u(t, \cdot; u_0)\|_\rho^2 \right) \leq C \mathcal{G}_\rho(t; u_0) < \infty. \quad (1.15) \text{ [E:uInLrho]}$$

Proof. Under condition (iii), we can apply Fubini's Theorem and the moment bound (3.6) below² to see that for some constant $C > 0$ independent of t , which may vary from line to line,

$$\begin{aligned} \mathbb{E} \left(\|u(t, \cdot; u_0)\|_\rho^2 \right) &= \mathbb{E} \left[\mathbb{E} \left(\int_{\mathbb{R}^d} |u(t, x; u_0)|^2 \rho(x) dx \middle| \mathcal{F}_0 \right) \right] \\ &\leq C \mathbb{E} \left[\int_{\mathbb{R}^d} \left(1 + (G(t, \cdot) * |u_0|)(x) \right)^2 \rho(x) dx \right] \end{aligned}$$

²For random initial condition, one only needs to take the expectation on both sides of (3.6).

$$\begin{aligned}
&= C \int_{\mathbb{R}^d} \mathbb{E} \left[\left(\left(G(t, \cdot) * (1 + |u_0|) \right)(x) \right)^2 \right] \rho(x) dx \\
&= C \mathcal{G}_\rho(t; u_0) < \infty.
\end{aligned}$$

This proves Theorem 1.2. $\square$

Here is the main theorem of the paper:

Theorem 1.3. *Let $u(t, x)$ be the solution to (1.1) starting from μ . Assume that all conditions in Theorem 1.2, with u_0 replaced by μ and ρ being an admissible weight (see Definition 3.1 below), are satisfied. Moreover, we assume that*

(i) *there exists another admissible weight $\widehat{\rho}$ such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\widehat{\rho}(x)} dx < \infty; \quad (1.16) \quad \boxed{\text{E:rho rho}}$$

(ii) *the weight function $\widehat{\rho}$ and the initial condition satisfy the following condition:*

$$\limsup_{t>0} \mathcal{G}_{\widehat{\rho}}(t; \mu) < \infty; \quad (1.17) \quad \boxed{\text{E:InitData}}$$

(iii) *for some $\alpha \in (128\Upsilon(0)L_b^2, 1/2)$ (see (1.10)), the spectral measure $\widehat{f}$ satisfies that*

$$\mathcal{H}_\alpha(t) := \int_0^t dr \, r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2) < \infty \quad \text{for all } t > 0. \quad (1.18) \quad \boxed{\text{E:H}}$$

Then we have that

(a) *for any $\tau > 0$, the sequence of laws of $\{\mathcal{L}u(t, \cdot; \mu)\}_{t \geq \tau}$ is tight, i.e., for any $\epsilon \in (0, 1)$, there exists a compact set $\mathcal{K} \subset L_\rho^2(\mathbb{R}^d)$ such that*

$$\mathcal{L}u(t, \cdot; \mu)(\mathcal{K}) := \mathbb{P}(u(t, \cdot; \mu) \in \mathcal{K}) \geq 1 - \epsilon, \quad \text{for all } t \geq \tau > 0; \quad (1.19) \quad \boxed{\text{E:cpt}}$$

(b) *there exists an invariant measure for the laws $\{\mathcal{L}u(t, \cdot; \mu)\}_{t>0}$ in $L_\rho^2(\mathbb{R}^d)$.*

We will postpone the proof of Theorem 1.3 (a) to Section 4. The proof of part (b) based on (a) is routine:

Proof of Theorem 1.3 (b). Fix an arbitrary $\tau > 0$ and denote

$$U(T) := \frac{1}{T} \int_\tau^{T+\tau} \mathcal{L}(u(t, \cdot; \mu)) \, dt, \quad T > 0.$$

We claim that the family of laws $U(T, \cdot)$ for $T > 0$ is tight in $L_\rho^2(\mathbb{R}^d)$. Indeed, for any $\epsilon \in (0, 1)$, by part (a), there exists a compact set $\mathcal{K} \subset L_\rho^2(\mathbb{R}^d)$ such that (1.19) holds. This implies that

$$U(T)(\mathcal{K}) = \frac{1}{T} \int_\tau^{T+\tau} \mathcal{L}(u(t, \cdot; \mu))(\mathcal{K}) \, dt \geq \frac{1}{T} \int_\tau^{T+\tau} (1 - \epsilon) \, dt = 1 - \epsilon, \quad \text{for all } T > 0.$$

Let $\{T_n\}_{n \in \mathbb{N}}$ be any deterministic sequence such that $T_n \uparrow \infty$. Since $\{U(T_n)\}_{n \geq 1}$ is a tight sequence of measures, there exists a subsequence $\{U(T_{n_m})\}_{m \geq 1}$ that converges weakly to a measure, say, η , on $L_\rho^2(\mathbb{R}^d)$. Then one can apply for the Krylov-Bogoliubov theorem (see, e.g., [DZ96, Theorem 3.1.1]) to conclude that the measure η is the invariant measure for $\mathcal{L}(u(t, \cdot; \mu))$, $t \geq \tau$. Finally, since τ can be arbitrarily close to zero, one can conclude part (b) of Theorem 1.3. $\square$

2 Discussions and Examples

2.1 Various initial conditions

Example 2.1 ($L^\infty(\mathbb{R}^d)$ initial condition). We emphasize that if the initial condition μ is deterministic and is such that $\mu(dx) = \varphi(x)dx$ with $\varphi \in L^\infty(\mathbb{R}^d)$, then all conditions related to $\mathcal{G}_\rho(\cdot)$ in both Theorems 1.2 and 1.3 are trivially satisfied. To be more precise, both conditions (1.14) and (1.17) hold because

$$\mathcal{G}_\rho(t; \varphi) \leq \|\varphi\|_{L^\infty(\mathbb{R}^d)}^2 \|\rho\|_{L^1(\mathbb{R}^d)} < \infty \quad \text{uniformly for all } t \geq 0.$$

Example 2.2 (Delta initial condition). In this example, we study the case when the initial condition μ is the Dirac delta measure at zero, namely, δ_0 . Let ρ be a nonnegative and $L^1(\mathbb{R}^d)$ function. Since

$$\mathcal{G}_\rho(t; \delta_0) = \int_{\mathbb{R}^d} G(t, x)^2 \rho(x) dx \leq G(t, 0)^2 \|\rho\|_{L^1(\mathbb{R}^d)}, \quad \text{for all } t > 0,$$

we see that both conditions (1.14) and (1.17) are satisfied. In particular, $\limsup_{t \rightarrow 0} \mathcal{G}_\rho(t; \delta_0) = 0$.

Example 2.3 (More initial conditions not in $L_\rho^2(\mathbb{R}^d)$). In this example, we study the case when $\mu(dx) = |x|^{-\alpha} dx$ for some $\alpha \in (0, d)$. It is clear that when $\alpha \in (d/2, d)$, $\mu \notin L_\rho^2(\mathbb{R}^d)$. However, in this case, we have

$$J_0(t, x) = (G(t, \cdot) * |\cdot|^{-\alpha})(x) \leq (G(t, \cdot) * |\cdot|^{-\alpha})(0).$$

Hence,

$$J_0(t, x) \leq C_d (2\pi t)^{-d/2} \int_0^\infty e^{-\frac{r^2}{2t}} r^{-\alpha+d-1} dr = C_* t^{-\alpha/2}, \quad \text{with } C_* = \frac{2^{-\alpha/2} \Gamma((d-\alpha)/2)}{\Gamma(d/2)},$$

which implies that

$$\mathcal{G}_\rho(t; |\cdot|^{-\alpha}) \leq \int_{\mathbb{R}^d} J_0^2(t, x) \rho(x) dx = C_*^2 t^{-\alpha} \|\rho\|_{L^1(\mathbb{R}^d)}, \quad \text{for all } t > 0.$$

Therefore, we see that both conditions (1.14) and (1.17) are satisfied.

The following proposition shows that for initial conditions with unbounded tails, condition (1.14) may hold while condition (1.17) may fail.

Proposition 2.4. Suppose that $\rho = \exp(-|x|)$, which is an admissible weight function. Let the initial condition μ be given as

$$\mu(dx) = |x|^\alpha dx \quad \text{with } \alpha > 0.$$

Then for some constants $C, C' > 0$, it holds that

$$C'(1+t^\alpha) \leq \mathcal{G}_\rho(t; \mu) \leq C(1+t^\alpha), \quad \text{for all } t > 0. \quad (2.1) \quad \boxed{\text{E:growtail}}$$

In particular, this implies that condition (1.14) is satisfied, but condition (1.17) fails.

Proof. Notice that by scaling arguments,

$$\mathcal{G}_\rho(t; \mu) = \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(t, x-y) |y|^\alpha dy \right)^2 e^{-|x|} dx$$

$$\begin{aligned}
&= \int_{\mathbb{R}^d} t^\alpha \left(\int_{\mathbb{R}^d} G \left(1, \frac{x}{\sqrt{t}} - z \right) |z|^\alpha dz \right)^2 e^{-|x|} dx \\
&= \int_{\mathbb{R}^d} t^{\alpha+d/2} \left(\int_{\mathbb{R}^d} G(1, \xi - z) |z|^\alpha dz \right)^2 e^{-\sqrt{t}|\xi|} d\xi.
\end{aligned}$$

In the following, let C , C_α and C'_α be generic constants that may depend on α and may change their value at each appearance.

Upper bound: Because

$$\int_{\mathbb{R}^d} G(1, z) |\xi - z|^\alpha dz \leq C_\alpha \int_{\mathbb{R}^d} G(1, z) (|\xi|^\alpha + |z|^\alpha) dz = C'_\alpha (1 + |\xi|^\alpha),$$

we see that

$$\begin{aligned}
\mathcal{G}_\rho(t; \mu) &\leq C_\alpha \int_{\mathbb{R}^d} t^{\alpha+d/2} (1 + |\xi|^{2\alpha}) e^{-\sqrt{t}|\xi|} d\xi \\
&= C'_\alpha \int_{\mathbb{R}} t^{\alpha+d/2} (1 + r^{2\alpha}) e^{-\sqrt{t}r} r^{d-1} dr \\
&= C'_\alpha (t^\alpha \Gamma(d) + \Gamma(d + 2\alpha)) \\
&= C_\alpha (1 + t^\alpha) < \infty.
\end{aligned}$$

Hence, this proves the upper bound in (2.1).

Lower bound: Now we prove the lower bound in (2.1). Indeed,

$$\begin{aligned}
\int_{\mathbb{R}^d} G(1, z) |\xi - z|^\alpha dz &\geq \int_{\mathbb{R}^d} G(1, z) ||\xi| - |z||^\alpha dz \\
&= \frac{C_d}{(2\pi)^{d/2}} \int_0^\infty ||\xi| - x|^\alpha e^{-\frac{x^2}{2}} x^{d-1} dx \\
&\geq C \int_1^2 ||\xi| - x|^\alpha dx \\
&= \frac{C}{1 + \alpha} \psi(|\xi|),
\end{aligned}$$

where, by consider three cases, we have

$$\begin{aligned}
\psi(r) &= \begin{cases} (2-r)^{\alpha+1} - (1-r)^{\alpha+1} & \text{if } 0 < r < 1, \\ (2-r)^{\alpha+1} + (r-1)^{\alpha+1} & \text{if } 1 \leq r \leq 2, \\ (r-1)^{\alpha+1} - (r-2)^{\alpha+1} & \text{if } r > 2, \end{cases} \\
&= \text{sgn}(2-r) |r-2|^{\alpha+1} + \text{sgn}(r-1) |r-1|^{\alpha+1}.
\end{aligned}$$

We claim that

$$\inf_{r \geq 0} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} > 0. \quad (2.2) \quad \boxed{\text{E:psi}}$$

With (2.2), we have that

$$\int_{\mathbb{R}^d} G(1, z) |\xi - z|^\alpha dz \geq C \sqrt{1 + |\xi|^{2\alpha}}.$$

Then, by the same arguments, we obtain the lower bound in (2.1). It remains to prove (2.2), which will be proved in three cases:

When $r > 2$, we see that

$$\begin{aligned} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} &\geq C_\alpha \frac{(r-1)^{\alpha+1} - (r-2)^{\alpha+1}}{(1+r)^\alpha} \\ &\geq C_\alpha \frac{(r-1)^\alpha(r-1) - (r-1)^\alpha(r-2)}{(1+r)^\alpha} \\ &= C_\alpha \left(\frac{r-1}{1+r} \right)^\alpha = C_\alpha \left(1 - \frac{2}{1+r} \right)^\alpha \geq C_\alpha \left(1 - \frac{2}{3} \right)^\alpha. \end{aligned}$$

which shows that $\inf_{r>2} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} > 0$.

When $r \in (1, 2]$, elementary calculations show that the minimum of $\psi(r)$ is achieved at $r = 3/2$. Hence,

$$\inf_{r \in (1, 2]} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} \geq \frac{\psi(3/2)}{\sqrt{1+4^\alpha}} > 0.$$

Similarly, when $r \in (0, 1]$, by differentiation, one finds that the function $\psi(r)$ is nonincreasing. Hence, the minimum is achieved at $r = 1$:

$$\inf_{r \in (0, 1]} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} \geq \frac{\psi(1)}{\sqrt{2}} > 0.$$

Combing the above three cases proves (2.2). This completes the proof of Proposition 2.4. $\square$

2.2 Bessel kernel and Matérn class of correlation functions

$\langle \text{Ex:Gamma+f} \rangle$

Example 2.5 (Bessel kernel as a correlation function). Let f_s denote the Bessel kernel with a strictly positive parameter $s > 0$. It is known that (see, e.g., Section 1.2.2 of [Gra14])

1. $f_s(x) > 0$ for all $x \in \mathbb{R}^d$ and $\|f_s\|_{L^1(\mathbb{R}^d)} = 1$;

2. there exists a constant $C(s, d) > 0$ such that

$$f_s(x) \leq C(s, d) \exp(-|x|/2) \quad \text{for } |x| \geq 2;$$

3. there exists a constant $c(s, d) > 0$ such that

$$\begin{aligned} \frac{1}{c(s, d)} &\leq \frac{f_s(x)}{H_s(x)} \leq c(s, d) \quad \text{for } |x| \leq 2, \quad \text{with} \\ H_s(x) &= \begin{cases} |x|^{s-d} + 1 + O(|x|^{s-d+2}) & \text{for } 0 < s < d, \\ \log\left(\frac{2}{|x|}\right) + 1 + O(|x|^2) & \text{for } s = d, \\ 1 + O(|x|^{s-d}) & \text{for } s > d; \end{cases} \end{aligned}$$

4. the Fourier transform of f_s is strictly positive:

$$\mathcal{F}f_s(\xi) = \frac{1}{(1 + |\xi|^2)^{s/2}}. \tag{2.3} \quad \boxed{\text{E:FB}}$$

Indeed, one can use (2.3) as the definition of the Bessel kernel. Properties 1 and 4 ensure that f_s is a nonnegative and nonnegative-definite tempered measure for all $s > 0$. From Property 4, we see that

$$\Upsilon(0) < \infty \iff s > d - 2.$$

In the following, we will assume that $s > d - 2$ and $d \geq 3$.

Example 2.6 (Matérn class of correlation functions). The *Matérn class of correlation functions* have been widely used in spatial statistics; one may check a recent work [LSW21] for references. Following Section 2.10 of [Ste99], this class of correlation functions are given by

$$K(x) = \phi \cdot (\alpha|x|)^\nu \mathcal{K}_\nu(\alpha|x|), \quad \text{for } x \in \mathbb{R}^d \text{ with } \phi > 0, \alpha > 0, \nu > 0, \quad (2.4) \quad \boxed{\text{E:Matern}}$$

where $\mathcal{K}_\nu(\cdot)$ is the modified Bessel function of second type, α and ν refer to the *scaling and smoothness parameters*, respectively. From the inversion formula (see p. 46 *ibid.*), one sees that

$$\mathcal{F}K(\xi) = (2\pi)^d \mathcal{F}^{-1}K(\xi) = (2\pi)^d f(|\xi|) \quad \text{with} \quad f(\xi) = \frac{2^{\nu-1} \phi \Gamma(\nu + d/2) \alpha^{2\nu}}{\pi^{d/2} (\alpha^2 + |\xi|^2)^{\nu+d/2}}, \quad \xi \in \mathbb{R}^d.$$

Comparing the above expression with (2.3), we see that the class of Bessel kernel f_s , with $s > d - 2$ and $d \geq 3$, includes the Matérn class (2.4) as a special case under the following choice of parameters:

$$\alpha = 1, \quad \nu = (s - d)/2, \quad \text{and} \quad \phi = 2^{(2-d-s)/2} \pi^{-d/2} \Gamma(s/2)^{-1}.$$

Note that the requirement of the smoothness parameter $\nu > 0$ for the Matérn class corresponds to the case of the Bessel kernel with $s > d$.

$\langle \text{P:Bessel} \rangle$ **Proposition 2.7.** *For the Bessel kernel $f_s(\cdot)$ with $s > 0$ defined in Example 2.5, it holds that*

$$\Upsilon(0) = \frac{2\Gamma((s-d+2)/2)}{\pi^{d/2}(d-2)\Gamma(s/2)}, \quad \text{for all } s > d - 2 \text{ and } d > 2, \text{ and} \quad (2.5) \quad \boxed{\text{E:UpsilonB}}$$

$$\mathcal{H}_\alpha(t) < \infty \quad \forall t > 0 \iff 0 < \alpha < \frac{1}{2} - \frac{(d-s)_+}{4} \quad \text{and} \quad s > d - 2 > 0, \quad (2.6) \quad \boxed{\text{E:HaB}}$$

where $a_+ := \max(a, 0)$. Moreover, for $\alpha \in (0, 1/2)$, we have the following series expansion of

$\mathcal{H}_\alpha(t)$ at $t \rightarrow 0$:

$$\mathcal{H}_\alpha(t) = \begin{cases} \frac{\pi^{d/2} \Gamma((d-s)/2)}{((s-d)/2 + 1 - 2\alpha) \Gamma(d/2)} t^{(s-d)/2 + 1 - 2\alpha} \\ - \frac{\pi^{d/2} \Gamma((s-d)/2)}{(1-2\alpha) \Gamma(s/2)} t^{1-2\alpha} \\ + O\left(t^{(s-d)/2 + 2(1-\alpha)}\right), & \begin{array}{l} d-2 < s < d \\ \text{and} \\ \alpha < \frac{1}{2} - \frac{1}{4}(d-s) \end{array} & (2.7\text{-a}) \\ \\ \frac{\pi^{d/2}}{(1-2\alpha) \Gamma(d/2)} t^{1-2\alpha} \log\left(\frac{1}{t}\right) \\ + \frac{\pi^{d/2} (1 - (1-2\alpha) [\psi(d/2) + 2\gamma])}{(1-2\alpha)^2 \Gamma(d/2)} t^{1-2\alpha} & s = d & (2.7\text{-b}) \\ + O(t^2 \log(t)), & & (2.7) \text{ [E:HaBS]} \\ \\ \frac{\pi^{d/2} \Gamma((s-d)/2)}{(1-2\alpha) \Gamma(s/2)} t^{1-2\alpha} + O(t^{(s-d)/2 + 1 - 2\alpha}), & d < s < d+2 & (2.7\text{-c}) \\ \\ \frac{\pi^{d/2}}{(1-2\alpha) \Gamma(d/2 + 1)} t^{1-2\alpha} + O(t^{2(1-\alpha)} \log(t)), & s = d+2 & (2.7\text{-d}) \\ \\ \frac{\pi^{d/2}}{(1-2\alpha) \Gamma(d/2 + 1)} t^{1-2\alpha} + O(t^{2(1-\alpha)}), & s > d+2 & (2.7\text{-e}) \end{cases}$$

where $\psi(x) = \frac{d}{dx} \log \Gamma(x)$ refers to the digamma function and $\gamma \approx 0.57721$ to Euler's constant; see, e.g., 5.2.2 and 5.2.3 on p. 136 of [AR10].

Proof. By the spherical coordinates and (2.3),

$$\Upsilon(0) = (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{d\xi}{|\xi|^2 (1 + |\xi|^2)^{s/2}} = (2\pi)^{-d} C_d \int_0^\infty \frac{r^{d-1}}{r^2 (1 + r^2)^{s/2}} dr, \quad C_d := \frac{2\pi^{d/2}}{\Gamma(d/2)}.$$

Now by change of variable $z = r^2/(1 + r^2)$, we can evaluate the above integral explicitly by transforming it to the Beta integral:

$$\int_0^\infty \frac{r^{d-1}}{r^2 (1 + r^2)^{s/2}} dr = \frac{1}{2} \int_0^1 z^{d/2-2} (1-z)^{(s-d)/2} dz = \frac{\Gamma(d/2 - 1) \Gamma((s-d+2)/2)}{2\Gamma(s/2)},$$

which is finite provided that $s > d - 2 > 0$. This proves (2.5).

It remains to prove (2.7), which then implies (2.6). From (1.18) and by the spherical coordinates, for all $t > 0$,

$$\begin{aligned} \mathcal{H}_\alpha(t) &= \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} d\xi \frac{\exp(-r|\xi|^2)}{(1 + |\xi|^2)^{s/2}} \\ &= C_d \int_0^t dr r^{-2\alpha} \int_0^\infty dz \frac{\exp(-rz^2)}{(1 + z^2)^{s/2}} z^{d-1} \\ &= \frac{C_d}{2} \int_0^t dr r^{-2\alpha} \int_0^\infty du \exp(-ru) (1+u)^{-s/2} u^{d/2-1} \end{aligned}$$

$$=: \frac{C_d \Gamma(d/2)}{2} \int_0^t dr r^{-2\alpha} I(r) = \pi^{d/2} \int_0^t dr r^{-2\alpha} I(r).$$

By [Old10, 13.4.4 on p.326], $I(r)$ is equal to the *confluent hypergeometric function*:

$$I(r) = U\left(\frac{d}{2}, \frac{2+d-s}{2}, r\right).$$

By 18.2.18 – 13.2.22 on p. 323 *ibid.*, we see that

$$I(r) = \begin{cases} \frac{\Gamma((d-s)/2)}{\Gamma(d/2)} r^{(s-d)/2} + \frac{\Gamma((s-d)/2)}{\Gamma(s/2)} + O(r^{(s-d)/2+1}) & d-2 < s < d & 18.2.18, \\ -\frac{1}{\Gamma(d/2)} (\log(r) + \psi(d/2) + 2\gamma) + O(r \log(r)) & s = d & 18.2.19, \\ \frac{\Gamma((s-d)/2)}{\Gamma(s/2)} + O(r^{(s-d)/2}) & d < s < d+2 & 18.2.20, \\ \frac{1}{\Gamma(d/2+1)} + O(r \log(r)) & s = d+2 & 18.2.21, \\ \frac{\Gamma((s-d)/2)}{\Gamma(s/2)} + O(r) & s > d+2 & 18.2.22. \end{cases}$$

Then integrating the right-hand side of the above expressions against $\pi^{d/2} r^{-2\alpha} dr$ over $[0, t]$ gives the five cases in (2.7). This completes the proof of Proposition 2.7. $\square$

2.3 Comparing Tessitore and Zabczyk's condition

$\langle \text{R:Fourier} \rangle$ **Remark 2.8.** The Fourier transform may be defined differently depending how to handle the 2π constant. In this paper (as in [CK19; CH19]), we use the convention that

$$\widehat{\phi}(\xi) = \mathcal{F}\phi(\xi) := \int_{\mathbb{R}^d} e^{-ix \cdot \xi} \phi(x) dx \quad \text{and} \quad \mathcal{F}^{-1}\psi(x) := (2\pi)^{-d} \int_{\mathbb{R}^d} e^{ix \cdot \xi} \psi(\xi) d\xi.$$

Hence, the Plancherel theorem takes the form of

$$\int_{\mathbb{R}^d} \psi(x) \phi(x) dx = (2\pi)^{-d} \int_{\mathbb{R}^d} \widehat{\psi}(\xi) \overline{\widehat{\phi}(\xi)} d\xi. \quad (2.8) \quad \boxed{\text{E:Plancherel}}$$

It has not been explicitly mentioned in [TZ98] about the convention of the Fourier transform. However, by checking the proof of Theorem 3.3 (*ibid.*) it suggests that the following convention has been used

$$\widehat{\phi}(\xi) = \mathcal{F}\phi(\xi) := (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-ix \cdot \xi} \phi(x) dx \quad \text{and} \quad \mathcal{F}^{-1}\psi(x) := (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{ix \cdot \xi} \psi(\xi) d\xi,$$

so that the Plancherel theorem takes the form of $\int_{\mathbb{R}^d} \psi(x) \phi(x) dx = \int_{\mathbb{R}^d} \widehat{\psi}(\xi) \overline{\widehat{\phi}(\xi)} d\xi$ without the additional factor $(2\pi)^{-d}$. When we translate their condition (3.4) (*ibid.*) to (1.12), we have taken into account of this difference. In particular, the spectral density γ (*ibid.*) corresponds $(2\pi)^{-d/2} \widehat{f}$ in this paper. This accounts the slightly different factor in front of the integral in (1.12) from that in (3.4) (*ibid.*).

Let us denote the right-side of the second condition in (1.12) by $M(f)$, namely,

$$M(f) := \Gamma(d/2 - 1) 2^{d/2-2} \int_{\mathbb{R}^d} \left(\left| \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right| * \left| \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right| \right) (\zeta) |\zeta|^{2-d} d\zeta,$$

and consider two cases:

Case I: If $\widehat{f}(\zeta)$ is strictly positive for all $\zeta \in \mathbb{R}^d$, then there is no ambiguity when taking square root and hence one can remove the absolute value to see that

$$\begin{aligned} M(f) &= \Gamma(d/2 - 1) 2^{d/2-2} \int_{\mathbb{R}^d} \left(\mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) * \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right) (\zeta) |\zeta|^{2-d} d\zeta \\ &= \frac{1}{(2\pi)^{3d/2}} \int_{\mathbb{R}^d} \widehat{f}(\zeta) |\zeta|^{-2} d\zeta = (2\pi)^{d/2} \Upsilon(0), \end{aligned}$$

where we have applied the Plancherel theorem (see (2.8)) and the Fourier transform for Riesz kernel (in the generalized sense):

$$\mathcal{F}(|\cdot|^{-\alpha})(\xi) = \pi^{-d/2} 2^{-\alpha} \frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)} |\xi|^{-(d-\alpha)}, \quad \text{for } \alpha \in (0, d) \text{ and } \xi \in \mathbb{R}^d.$$

Hence, the second condition in (1.12) can be equivalently written as $L_b^{-2} > (2\pi)^{d/2} \Upsilon(0)$. Comparing this with the second condition in (1.8), namely, $L_b^{-2} > 256 \Upsilon(0)$, our condition is sharper when $d > \frac{16 \log(2)}{\log(2\pi)} \approx 6.03$.

Case II: If $\widehat{f}(\cdot)$ is not strictly positive, but only nonnegative, namely, $\widehat{f}(\xi) = 0$ for some $\xi \in \mathbb{R}^d$, how to remove the absolute value in (1.12) becomes tricky, which will be illustrated in the following example:

Example 2.9. Let $d = 1$ and $g(x) = \frac{1}{2} 1_{[-1,1]}(x)$. It is clear that $\widehat{g}(\xi) = \xi^{-1} \sin(\xi)$. Now set $f(x) = (g * g)(x) = 2^{-2} \max(2 - x, 0)$. It is clear that f is nonnegative. It is also nonnegative-definite because $\widehat{f}(\xi) = \widehat{g}(\xi)^2 = \xi^{-2} \sin^2(\xi) \geq 0$. But in this case, $\widehat{f}(\cdot)$ is only nonnegative (not strictly positive) with infinite many zeros. Hence, when taking the square root of $\widehat{f}(\xi)$ as in (1.12), one needs to wisely choose the correct positive and negative branches:

1. Clearly, the signed version $\sqrt{\widehat{f}(\xi)} = \xi^{-1} \sin(\xi)$ is preferable since its inverse Fourier transform can be easily computed, which is equal to $g(x)$. Moreover, because this inverse Fourier transform $g(x)$ is clearly nonnegative, the absolute value signs in (1.12) do not pose any additional restrictions.
2. However, if one chooses the positive branches, namely, $\sqrt{\widehat{f}(\xi)} = |\xi^{-1} \sin(\xi)|$, it is not clear how to compute its Fourier transform. In general, some bad choices of the positive/negative branches may make the conditions in (1.12) fail. For example, such choice may turn $\sqrt{\widehat{f}(\xi)}$ into a distribution, then taking the absolute value of a distribution (unless it is a measure) may be problematic. Or even one may find some choice such that $\sqrt{\widehat{f}(\xi)}$ is a well-defined function, it may well be possible to be signed and after taking absolute values, the integral in (1.12) may blow up.

3 Preliminaries

(S:Prelim) Let ρ be some nonnegative function on $\mathbb{R}^d$. We denote by $L_\rho^2(\mathbb{R}^d)$ the weighted Hilbert space with the inner produce $\langle f, g \rangle_\rho := \int_{\mathbb{R}^d} f(x)g(x)\rho(x) dx$ and set $\|f\|_\rho^2 := \int_{\mathbb{R}^d} f(x)^2\rho(x) dx$. Following Tessitore and Zabczyk [TZ98], we make the following definition:

$\langle \text{D:Rho} \rangle$ **Definition 3.1.** $\rho : \mathbb{R}^d \mapsto \mathbb{R}$ is called an *admissible weight* if ρ is a strictly positive, bounded and continuous function in $L^1(\mathbb{R}^d)$ such that for $T > 0$ there exists a constant $C_\rho(T)$ such that

$$(G(t, \cdot) * \rho(\cdot))(x) \leq C_\rho(T)\rho(x) \quad \text{for all } t \in [0, T] \text{ and } x \in \mathbb{R}^d, \quad (3.1) \quad \boxed{\text{E:aw}}$$

or equivalently,

$$\sup_{x \in \mathbb{R}^d, t \in [0, T]} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq C_\rho(T). \quad (3.2) \quad \boxed{\text{E:awbd1?}}$$

Here are several examples of admissible weights (see Example in Section 2 of [TZ98])

$$\begin{aligned} \rho(x) &= \exp(-a|x|), & a > 0 \quad \text{and} \\ \rho(x) &= (1 + |x|^a)^{-1}, & a > d. \end{aligned} \quad (3.3) \quad \boxed{\text{E:admRho}}$$

$\langle \text{R:admRho} \rangle$ **Remark 3.2.** The smaller the weight function $\rho(\cdot)$ (not necessarily to be admissible here) is, the larger the space $L_\rho^2(\mathbb{R}^d)$ is. For example, one may choose ρ to be either a nonnegative function with compact support or the heat kernel itself $G(1, \cdot)$. In both cases, ρ is smaller than those in (3.3) (up to a constant). However, one can easily check that the admissible condition (3.1) excludes these two cases. It is much less obvious that the following weight function is not admissible:

$$\rho(x) = \exp(-a|x|^b), \quad x \in \mathbb{R}^d, \quad \text{with } a > 0 \text{ and } b \in (1, 2) \text{ fixed.}$$

$\langle \text{P:weight} \rangle$ **Proposition 3.3** (Proposition 2.1 of [TZ98]). *For any admissible weight ρ , the operators on $L_\rho^2(\mathbb{R}^d)$ defined by $\varphi \mapsto (G(t, \cdot) * \varphi(\cdot))(x)$ can be extended to a C_0 -semigroup on L_ρ^2 . Moreover, if $\widehat{\rho}$ is another admissible weight such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\widehat{\rho}(x)} dx < \infty,$$

then for any $t > 0$, the operators defined above are compact from $L_\rho^2(\mathbb{R}^d)$ to $L_{\widehat{\rho}}^2(\mathbb{R}^d)$.

$\langle \text{T:Solve} \rangle$ **Theorem 3.4** (Theorem 1.2 [CH19]). *Suppose that*

- (i) *the initial measure μ is an $\mathcal{F}_0$ -measurable random variable that satisfies the following integrability condition:*

$$\mathbb{E} \left(\int_{\mathbb{R}^d} \exp(-a|x|^2) |\mu|(dx) \right) < \infty \quad \text{for all } a > 0, \quad (3.4) \quad \boxed{\text{A:icon}}$$

- (ii) *the spectral measure $\widehat{f}$ satisfies Dalang's condition (1.9),*

Then (1.1) has a unique random field solution starting from μ . Moreover, the solution is $L^2(\Omega)$ continuous and is adapted to the filtration $\{\mathcal{F}_t\}_{t \geq 0}$.

$\langle \text{T:ExUn} \rangle$ **Theorem 3.5** (Theorem 1.7 [CH19]). *Under the assumptions of Theorem 3.4, for any $t > 0$, $x \in \mathbb{R}^d$ and $p \geq 2$, the solution to (1.1), $u(t, x)$, given by (1.3) is in $L^p(\Omega)$ and*

$$\|u(t, x)\|_p \leq [\bar{\varsigma} + \sqrt{2}(G(t, \cdot) * |\mu|)(x)] H(t; \gamma_p)^{1/2}, \quad (3.5) \quad \boxed{\text{E:mb}}$$

where $\bar{\varsigma} = L_0/L_b$, $\gamma_p = 32pL_b^2$ (see (1.4) for L_0 and L_b) and the function $H(t; \gamma_p)$ is nondecreasing in t (see [CH19] for the expression of the function H).

Corollary 3.6. *Under the same setting as Theorem 3.5, if the two conditions in (1.8) hold (see also (1.10)), then*

$$\|u(t, x)\|_p \leq C_p \left(1 + (G(t, \cdot) * |\mu|)(x) \right), \quad \text{for all } p \text{ s.t. } 1/p \in (128L_b^2\Upsilon(0), 1/2), \quad (3.6) \text{ [E:mb0]}$$

where $C_p = (\sqrt{2} \vee \bar{\varsigma}) \sup_{t \geq 0} H(t; \gamma_p)^{1/2} < \infty$.

Proof. Lemma 2.5 of [CK19] gives one sufficient condition, namely, $2\gamma_p\Upsilon(0) < 1$, under which the function $H(t; \gamma_p)$ is bounded in t . Therefore, by taking into account the expression of γ_p in Theorem 3.5, we see that as a direct consequence of (3.5), whenever

$$32pL_b^2 < \frac{1}{4\Upsilon(0)}, \quad (3.7) \text{ [E:pLip]}$$

we have all p -th moments ($p \geq 2$) bounded as given in (3.6). $\square$

Remark 3.7 (Restarted SHE). Recall that the Markov property of the solution to (1.1) implies that for any $t \geq t_0 > 0$,

$$u(t + t_0, x; \mu) \stackrel{\mathcal{L}}{=} u(t, x; u(t_0, \cdot; \mu)) =: v(t, x), \quad (3.8) \text{ [E:eqLaw]}$$

where $\mathcal{L}$ refers to the equality in law and we have used $v(t, x)$ to simplify the notation. It is known that v satisfies the following restarted SPDE:

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2} \Delta v(t, x) = b(x, v(t, x)) \dot{W}_{t_0}(t, x) & x \in \mathbb{R}^d, t > 0, \\ v(0, x) = u(t_0, x; \mu), & x \in \mathbb{R}^d, \end{cases} \quad (3.9) \text{ [E:restart]}$$

where $\dot{W}_{t_0}(t, x) := \dot{W}(t + t_0, x)$ denotes the time shifted noise, i.e.,

$$\int_0^t \int_{\mathbb{R}^d} W_{t_0}(ds, dy) = \int_{t_0}^{t+t_0} \int_{\mathbb{R}^d} W(ds, dy). \quad (3.10) \text{ [E:ShiftW]}$$

Under conditions in (1.8), Theorem 3.5 and (3.8) imply immediately that

$$\|v(t, x)\|_q = \|u(t + t_0, x; \mu)\|_q \leq C_q \left(1 + (G(t + t_0, \cdot) * |\mu|)(x) \right) = C_q J_0(t + t_0, x; \mu^*), \quad (3.11) \text{ [E:MomBddRst]}$$

for all $q \geq 2$ and $t > 0$, where the constant C_q does not depend on t . Moreover, under the assumptions of Theorem 1.2, we have $v(0, \cdot) \in L_\rho^2(\mathbb{R}^d)$ a.s. and

$$\mathbb{E} \left(\|v(t, x)\|_\rho^2 \right) = \mathbb{E} \left(\|u(t + t_0, x; \mu)\|_\rho^2 \right) \leq C \mathcal{G}_\rho(t + t_0; \mu) < \infty. \quad (3.12) \text{ [E:MomRhoBddRst]}$$

This is a very import observation. Recall that Tessitore et.al. [TZ98, Theorem 3.3] proved that there exists an invariant measure, η , on $\mathcal{B}(L_\rho^2(\mathbb{R}^d))$ of the form (1.6) for the constant initial data $\varphi = 1$. Because of the restart property of the equation, we may rewrite (1.6) as the following:

$$\begin{aligned} \eta(A) &= \lim_{n \rightarrow \infty} \int_{t_0}^{T_n + t_0} \mathcal{L}(u(t, \cdot, 1))(A) dt \\ &= \lim_{n \rightarrow \infty} \int_{t_0}^{T_n + t_0} \mathcal{L} \left[v \left(t - \frac{t_0}{2}, \cdot, u \left(\frac{t_0}{2}, \cdot, 1 \right) \right) \right] (A) dt \end{aligned}$$

$$= \lim_{n \rightarrow \infty} \int_{t'_0}^{T_n+t'_0} \mathcal{L} \left[v \left(t, \cdot, u \left(\frac{t_0}{2}, \cdot, 1 \right) \right) \right] (A) dt, \quad t'_0 := t_0/2,$$

where we recall that $A \in \mathcal{B}(L^2_\rho(\mathbb{R}^d))$ and v denotes the solution to (3.9) starting from $u(t_0/2, \cdot, 1)$ with the time-shifted noise $\dot{W}_{t_0/2}$. Naturally the invariant measure is usually defined on the Hilbert space of functions where the initial condition lives; in the case of [TZ98], this is $L^2_\rho(\mathbb{R}^d)$. The above shows that we can extend the measure η to $\mathcal{B}(E_{t_0})$ since we can associate the equation (1.1) starting from $\mu = 1$ with the above restarted process v that starts from $u(t_0, \cdot, 1) \in E_{t_0}$. This is no contradiction since clearly $\mathcal{B}(L^2_\rho(\mathbb{R}^d)) \subset \mathcal{B}(E_{t_0})$.

This observation will allow us to show existence of an invariant measure in $\mathcal{B}(E_{t_0})$ for our equation (1.1) starting from an initial condition outside of $L^2_\rho(\mathbb{R}^d)$. For example, we will see in Example 2.5 below that we may start our equation at the Dirac delta measure δ_0 , which is clearly not in $L^2_\rho(\mathbb{R}^d)$. However, because of the smoothing properties of the heat operator, after any amount of time $t_0 > 0$, the solution $u(t_0, \cdot, \delta_0) \in E_{t_0}$. Thus, whether we originally start our equation (1.1) at the constant 1, as in [TZ98], or at a distribution that satisfies the assumptions of Theorem 1.3 main below, like the Dirac delta distribution δ_0 , then either way we end up in the same function space. If one were able to prove uniqueness of the invariant measure, then the invariant measure which is derived from the Dirac delta initial condition and the invariant measure which is derived from the constant 1 initial condition would be the same.

The following factorization property of the heat kernel will be used in this paper

$$G(t, x)G(s, y) = G\left(\frac{ts}{t+s}, \frac{sx+ty}{t+s}\right) G(t+s, x-y), \quad (3.13) \quad \boxed{\text{E:gsim}}$$

which can be easily verified.

4 Proof of part (a) of Theorem 1.3

(S:mainproof) Now, for $\alpha \in (0, 1)$, $t > 0$ and $x \in \mathbb{R}^d$, define formally

$$\begin{aligned} (F_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t-s)^{\alpha-1} G(t-s, x-y) f(s, y) ds dy \quad \text{and} \\ (Y_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t-s)^{-\alpha} G(t-s, x-y) f(s, y) W(ds, dy). \end{aligned} \quad (4.1) \quad \boxed{\text{E:Ops?}}$$

For F_α , the Step 1 of the proof of [TZ98, Theorem 3.1] showed the following proposition:

(P:F-Comp) **Proposition 4.1.** *Let ρ and $\hat{\rho}$ be given in condition (i) of Theorem 1.3 (see (1.16)). For any $q > 2$, $t_0 > 0$ and $\alpha \in (q^{-1}, 2^{-1})$, the operator F_α as an operator from $L^q((0, t_0); L^2_\rho(\mathbb{R}^d))$ to $L^2_\rho(\mathbb{R}^d)$ is compact.*

As for Y_α , we have the following two lemmas, which hold for both non-restarted SHE ($t_0 = 0$) and the restarted SHE ($t_0 > 0$):

(L:yineq) **Lemma 4.2.** *Suppose that μ — the initial condition for u — satisfies (3.4) and $\hat{f}$ satisfies Dalang's condition (1.9). Assume that the integral in (1.18) is finite for some $\alpha \in (0, 1/2)$. Fix an arbitrary $t_0 \geq 0$. Let $v(t, x)$ be the solution to the restarted SHE (3.9) and $\dot{W}_{t_0}$ be the time-shifted noise (see (3.10)) when $t_0 > 0$ and let $v = u$ when $t_0 = 0$. Then*

$$Y_v(s, y) := [Y_\alpha b(\circ, v(\cdot, \circ))](s, y) = \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, v(r, z)) W_{t_0}(dr, dz) \quad (4.2) \quad \boxed{\text{E:Y?}}$$

has the following properties:

(1) For all $q \geq 2$, $s > 0$, $y \in \mathbb{R}^d$,

$$\|Y_v(s, y)\|_q^2 \leq H(s + t_0; 32qL_b^2) J_0^2(s + t_0, y; \mu^*) \mathcal{H}_\alpha(s) < \infty, \quad (4.3) \quad \boxed{\text{E:NormY}}$$

where we refer to Theorem 3.5 for the function $H(t; \gamma)$;

(2) Under (1.8), if the integral in (1.18) is finite for some $\alpha \in (128L_b^2\Upsilon(0), 1/2)$, then for any q with $1/q \in (128L_b^2\Upsilon(0), \alpha)$, the function $H(t; 32qL_b^2)$ in (4.3) is uniformly bounded in $t \geq 0$, i.e., $\sup_{t \geq 0} H(t; 32qL_b^2) < \infty$;

(3) Under (1.8), if the integral in (1.18) is finite for some $\alpha \in (128L_b^2\Upsilon(0), 1/2)$, then for any q with $1/q \in (128L_b^2\Upsilon(0), \alpha)$ and for any nonnegative and $L^1(\mathbb{R}^d)$ -function ρ , there exists a constant $\Theta = \Theta(q, L_b, L_0, \alpha)$, which does not depend on t , such that for $t > 0$,

$$\mathbb{E} \left(\int_0^t \|Y_v(s, \cdot)\|_\rho^q ds \right) \leq \Theta \int_0^t [\mathcal{G}_\rho(s + t_0; \mu) \mathcal{H}_\alpha(s)]^{q/2} ds, \quad (4.4) \quad \boxed{\text{E:yineq}}$$

which is finite provided that

$$\int_0^t [\mathcal{G}_\rho(s + t_0; \mu) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty. \quad (4.5) \quad \boxed{\text{E:GH-int'}}$$

Remark 4.3. Condition (4.5) is true for $t_0 > 0$ because $\mathcal{G}_\rho(t; \mu)$ is a continuous function for $t > 0$ and $\mathcal{H}_\alpha(s)$ is continuous and bounded for $s \in [0, t]$ thanks to (1.18). However, when $t_0 = 0$, the situation is much more trickier. For example, when the initial condition is the delta initial condition, we have that

$$\mathcal{G}_{\hat{\rho}}(t; \delta_0) = \int_{\mathbb{R}^d} G(t, x)^2 \rho(x) dx = G(2t, 0) \int_{\mathbb{R}^d} G(t/2, x) \rho(x) dx < \infty,$$

where one can obtain the second equality via (3.13). Hence, when $s \rightarrow 0$, $\mathcal{G}_{\hat{\rho}}(s; \delta_0)$ blows up with a rate $s^{-d/2}$. On the other hand, $\mathcal{H}_\alpha(s)$ goes to zero with a different rate. One needs to combine these two rates to check if condition (4.5) holds. By introducing t_0 (or restart the heat equation), one can avoid this tricky issue.

Proof. In the proof, we use C to denote a generic constant that may change its value at each appearance.

(1) We first prove (4.3). By the Burkholder-Davis-Gundy inequality and Minkowski's integral inequality, we see that

$$\begin{aligned} \|Y_v(s, y)\|_q^2 &\leq C \int_0^s dr (s - r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) \|b(z_1, v(r, z_1))\|_q \\ &\quad \times f(z_1 - z_2) G(s - r, y - z_2) \|b(z_2, v(r, z_2))\|_q. \end{aligned}$$

Note that for the Lipschitz condition in (1.4), we have that

$$|b(x, v)| \leq |b(x, v) - b(x, 0)| + |b(x, 0)| \leq L_b |v| + L_0 \leq C(1 + |v|), \quad C := L_b \vee L_0.$$

We apply this and the moment bound (3.5) to $\|b(z_i, v(r, z_i))\|_q$ above to see that

$$\begin{aligned} \|b(z_i, v(r, z_i))\|_q &\leq C \left(1 + \|v(r, z_i)\|_q \right) \\ &= C \left(1 + \|u(r + t_0, z_i)\|_q \right) \end{aligned}$$

$$\begin{aligned}
&\leq CH(r+t_0; 32qL_b^2) J_0(r+t_0, z_i; \mu^*) \\
&\leq CH(s+t_0; 32qL_b^2) J_0(r+t_0, z_i; \mu^*), \quad i=1, 2, r \in (0, s),
\end{aligned} \tag{4.6} \quad \boxed{\text{E:NormB}}$$

where in the last step, we have used the fact that $H(t; \gamma)$ is a nondecreasing function; see Lemma 2.6 of [CK19]. Therefore, by denoting $C_s := H(s+t_0; 32qL_b^2)$,

$$\begin{aligned}
\|Y_v(s, y)\|_q^2 &\leq CC_s \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(s-r, y-z_i) J_0(r+t_0, z_i; \mu^*) \right) \\
&= CC_s \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu^*(d\sigma_1) \mu^*(d\sigma_2) \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\
&\quad \times f(z_1 - z_2) \prod_{i=1}^2 \left(G(s-r, y-z_i) G(r+t_0, z_i - \sigma_i) \right) \\
&= CC_s \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu^*(d\sigma_1) \mu^*(d\sigma_2) G(s+t_0, y - \sigma_1) G(s+t_0, y - \sigma_2) \\
&\quad \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 G\left(\frac{(r+t_0)(s-r)}{s+t_0}, z_i - \sigma_i \frac{r+t_0}{s+t_0} - \frac{s-r}{s+t_0} y\right) \\
&\leq CC_s (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu^*(d\sigma_1) \mu^*(d\sigma_2) G(s+t_0, y - \sigma_1) G(s+t_0, y - \sigma_2) \\
&\quad \times \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{(r+t_0)(s-r)}{s+t_0} |\xi|^2\right),
\end{aligned}$$

where we have applied (3.13) and Plancherel's theorem. Hence,

$$\|Y_v(s, y)\|_q^2 \leq CC_s (2\pi)^{-2d} J_0^2(s+t_0, y; \mu^*) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{(r+t_0)(s-r)}{s+t_0} |\xi|^2\right).$$

Because the function

$$t_0 \mapsto \frac{r+t_0}{s+t_0} = 1 - \frac{s-r}{s+t_0} \quad \text{for } t_0 > 0,$$

is nondecreasing in t_0 whenever $s > r > 0$, we can replace the two appearances of t_0 in the exponent of the above inequality by zero to see that

$$\|Y_v(s, y)\|_q^2 \leq CC_s (2\pi)^{-2d} J_0^2(s+t_0, y; \mu^*) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right). \tag{4.7} \quad \boxed{\text{E_:NormY}}$$

Furthermore, by symmetry of $r(s-r)/s$ and the fact that $r(s-r)/s \geq r/2$ for all $r \in [0, s/2]$, we see that the above double integral is bounded by

$$\begin{aligned}
&\leq 2 \int_0^{s/2} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r}{2} |\xi|^2\right) \\
&= 2^{2(1-\alpha)} \int_0^{s/4} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r |\xi|^2) \\
&\leq 2^{2(1-\alpha)} \int_0^s dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r |\xi|^2) \\
&= 2^{2(1-\alpha)} \mathcal{H}_\alpha(s).
\end{aligned}$$

Plugging the above bound back to (4.7) proves (4.3).

(2–3) Part (2) is a direct consequence of Theorem 3.5. It remains to prove (4.4). An application of Minkowski's inequality shows that

$$\mathbb{E} \left(\|Y_v(s, \cdot)\|_\rho^q \right) = \left\| \int_{\mathbb{R}^d} Y_v(s, y)^2 \rho(y) dy \right\|_{q/2}^{q/2} \leq \left(\int_{\mathbb{R}^d} \|Y_v(s, y)\|_q^2 \rho(y) dy \right)^{q/2}. \quad (4.8) \quad \boxed{\text{E:EYqr}}$$

By the definition of $\mathcal{G}_\rho(t; \mu)$ in (1.14) and by (4.3), we see that

$$\int_{\mathbb{R}^d} \|Y_v(s, y)\|_q^2 \rho(y) dy \leq C \mathcal{G}_\rho(s + t_0; \mu) \mathcal{H}_\alpha(s).$$

Plugging the above expression to the far right-hand of (4.8) proves (4.4). Finally, the finiteness of the upper bound in (4.4) is guaranteed by condition (4.5). This completes the proof of Lemma 4.2. $\square$

(L:Factorize)

Lemma 4.4 (Factorization lemma). *Suppose that μ — the initial condition for u — satisfies (3.4) and $\hat{f}$ satisfies Dalang's condition (1.9). Assume that condition (1.18) is satisfied for some $\alpha \in (0, 1/2)$. Fix an arbitrary $t_0 \geq 0$. Let $v(t, x)$ be the solution to the restarted SHE (3.9) and $\tilde{W}_{t_0}$ be the time-shifted noise (see (3.10)) when $t_0 > 0$ and let $v = u$ when $t_0 = 0$. Then the following factorization holds*

$$\frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y_v(s, \cdot)](x) ds = \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(z, v(r, z)) W_{t_0}(dr, dz),$$

for all $t > 0$ and $x \in \mathbb{R}^d$. As a consequence,

$$v(t, x) = [G(t, \cdot) * u(t_0, \cdot; \mu)](x) + \frac{\sin(\alpha\pi)}{\pi} [F_\alpha Y_v](t, x), \quad \text{for all } t > 0 \text{ and } x \in \mathbb{R}^d. \quad (4.9) \quad \boxed{\text{E:1fac?}}$$

Proof. The lemma is straightforward provided that one can switch the orders of stochastic and ordinary integrals:

$$\begin{aligned} & \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y_v(s, \cdot)](x) ds \\ &= \int_0^t ds (t-s)^{\alpha-1} \int_{\mathbb{R}^d} dy G(t-s, x-y) \\ & \quad \times \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, v(r, z)) W_{t_0}(dr, dz) \\ &= \int_0^t ds (t-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(t-r, x-z) b(z, v(r, z)) W_{t_0}(dr, dz) \end{aligned} \quad (4.10) \quad \boxed{\text{E:Fubini}_1}$$

$$\begin{aligned} &= \int_0^t \int_{\mathbb{R}^d} W(dr, dz) G(t-r, x-z) b(z, v(r, z)) \int_r^t ds (s-r)^{-\alpha} (t-s)^{\alpha-1} \\ &= \frac{\pi}{\sin(\alpha\pi)} \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(z, v(r, z)) W_{t_0}(dr, dz), \end{aligned} \quad (4.11) \quad \boxed{\text{E:Fubini}_2}$$

where the last step is the *Beta integral* which requires that $\alpha \in (0, 1)$. It remains to justify the two applications of the stochastic Fubini's theorem (see Theorem 5.30 of Chapter one in [Dal+09], or also [Wal86] or Theorem 4.18 of [DZ92]) in (4.10) and (4.11) in the following two steps.

Step 1. In this step, we justify the change of orders in (4.10). Note that t, x and s are fixed. It suffices to prove the following condition:

$$I_1 := \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2$$

$$\begin{aligned}
& \times f(z_1 - z_2) \left(\prod_{i=1}^2 G(s - r, y - z_i) \right) \mathbb{E} \left(\prod_{i=1}^2 b(z_i, v(r, z_i)) \right) \\
& = \int_{\mathbb{R}^d} dy G(t - s, x - y) \|Y_v(s, y)\|_2^2 \\
& < +\infty.
\end{aligned}$$

But this follows immediately from (4.3). Indeed

$$\begin{aligned}
\int_{\mathbb{R}^d} dy G(t - s, x - y) \|Y_v(s, y)\|_2^2 & \leq C \int_{\mathbb{R}^d} dy G(t - s, x - y) J_0^2(s + t_0, y; \mu^*) \mathcal{H}_\alpha(s) \\
& = C \mathcal{H}_\alpha(s) \int_{\mathbb{R}^d} dy G(t - s, x - y) \iint_{\mathbb{R}^{2d}} \mu^*(dz_1) \mu^*(dz_2) \\
& \quad \times G(s + t_0, y - z_1) G(s + t_0, y - z_2).
\end{aligned}$$

Now we bound the three heat kernels using (3.13) as follows:

$$\begin{aligned}
& G(t - s, x - y) \prod_{i=1}^2 G(s + t_0, y - z_i) \\
& = \frac{G(2(t - s), x - y)^2}{G(4(t - s), 0)} \prod_{i=1}^2 G(s + t_0, y - z_i) \\
& \leq 2^d \frac{G(2(t - s), x - y)^2}{G(4(t - s), 0)} \prod_{i=1}^2 G(2s + 2t_0, y - z_i) \\
& = 2^d [4(t - s)]^{d/2} \prod_{i=1}^2 \left[G(2s + 2t_0, y - z_i) G(2(t - s), x - y) \right] \\
& = 2^{2d} (t - s)^{d/2} \prod_{i=1}^2 \left[G(2(t + t_0), x - z_i) G\left(\frac{2(t - s)(s + t_0)}{t + t_0}, y - \frac{s + t_0}{t + t_0}(x - z_i)\right) \right] \\
& \leq 2^{2d} (t - s)^{d/2} \prod_{i=1}^2 \left[G(2(t + t_0), x - z_i) G\left(\frac{2(t - s)(s + t_0)}{t + t_0}, 0\right) \right] \\
& \leq C_{t,s,t_0} \prod_{i=1}^2 G(2(t + t_0), x - z_i).
\end{aligned}$$

Therefore, $I_1 \leq C_{t,s,t_0} \mathcal{H}_\alpha(s) J_0^2(2(t + t_0), x; \mu^*) < \infty$.

Step 2. Similarly, as for (4.11), we need to show that

$$\begin{aligned}
I_2 & := \int_0^t ds (t - s)^{\alpha-1} \int_0^s dr (s - r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\
& \quad \times f(z_1 - z_2) \left(\prod_{i=1}^2 G(t - r, x - z_i) \right) \mathbb{E} \left(\prod_{i=1}^2 b(z_i, v(r, z_i)) \right) < \infty.
\end{aligned}$$

By the Cauchy Schwartz inequality, (4.6) and because $\alpha \in (0, 1/2)$,

$$I_2 \leq C \int_0^t ds (t - s)^{\alpha-1} \int_0^s dr (s - r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(t - r, x - z_i) J_0(r + t_0, z_i; \mu^*) \right)$$

$$= C' \int_0^t dr (t-r)^{-\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(t-r, x - z_i) J_0(r + t_0, z_i; \mu^*) \right).$$

Now by the same arguments as those leading to (4.3) (with 2α there replaced by α), we see that

$$I_2 \leq C J_0^2(t, x; \mu^*) \int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t} |\xi|^2\right),$$

which is finite by (1.18) where we replace α with $\alpha/2$ and repeat the same steps right after (4.7). This completes the proof of Lemma 4.4. $\square$

We are now ready to prove part (a) of Theorem 1.3.

Proof of Theorem 1.3 (a). In this proof, $u(t, x)$ refers to $u(t, x; \mu)$. Fix $\tau > 0$ and $t_0 = \tau/2$. Throughout the proof, we have $t \geq \tau$. Let v be the solution to (3.9) that is restarted from $t - t_0$. Then

$$v(t_0, \cdot) \stackrel{\mathcal{L}}{=} u(t, \cdot) \quad \text{for } t \geq \tau.$$

According to Assumption (i), we can choose and fix some admissible weight function $\widehat{\rho}$ such that (1.16) is satisfied. Hence, by Proposition 3.3, the following set

$$\mathcal{K}_1(\Lambda) := \left\{ (G(t_0, \cdot) * y(\cdot))(x) : \|y\|_{\widehat{\rho}} \leq \Lambda \right\} \quad \text{with } \Lambda > 0$$

is relative compact in $L_{\widehat{\rho}}^2(\mathbb{R}^d)$. Thanks to (1.8) (see also (1.10)) and Assumption (iii), we can choose and fix some q and α with

$$128L_b^2\Upsilon(0) < \frac{1}{q} < \alpha < \frac{1}{2} \quad \left(\text{or equivalently } 2 < \frac{1}{\alpha} < q < \frac{1}{128L_b^2\Upsilon(0)} \right) \quad (4.12) \quad \boxed{\text{E:Choice}}$$

such that (1.18) holds. Hence, by Lemma 4.4, we have the following factorization:

$$v(t_0, x) = \left(G(t_0, \cdot) * u(t - t_0, \cdot) \right)(x) + \frac{\sin(\alpha\pi)}{\pi} [F_{\alpha} Y_v](t_0, x). \quad (4.13) \quad \boxed{\text{E:Fact}}$$

In particular, the operator F_{α} is well defined. With the choice of α and q in (4.12), we can apply Proposition 4.1 to see that the following set

$$\mathcal{K}_2(\Lambda) := \left\{ (F_{\alpha} h)(x) : \|h\|_{L^q((0, t_0); L_{\widehat{\rho}}^2(\mathbb{R}^d))} \leq \Lambda \right\} \quad \text{with } \Lambda > 0$$

is relative compact in $L_{\widehat{\rho}}^2(\mathbb{R}^d)$. Now for any $\Lambda > 0$, define the set $\mathcal{K}(\Lambda)$ as

$$\begin{aligned} \mathcal{K}(\Lambda) &:= \mathcal{K}_1(\Lambda) + \mathcal{K}_2(\Lambda) \\ &= \left\{ (G(t_0, \cdot) * y(\cdot))(x) + (F_{\alpha} h)(x) : \|y\|_{\widehat{\rho}} \leq \Lambda \quad \text{and} \quad \|h\|_{L^q((0, t_0); L_{\widehat{\rho}}^2(\mathbb{R}^d))} \leq \Lambda \right\}. \end{aligned}$$

Notice that from the factorization formula (4.13),

$$\begin{aligned} \mathbb{P}[v(t_0, \cdot) \notin \mathcal{K}(\Lambda)] &\leq \mathbb{P} \left[\left(\int_0^{t_0} \|Y_v(s, \cdot)\|_{\widehat{\rho}}^q ds \right)^{1/q} > \frac{\pi\Lambda}{\sin(\alpha\pi)} \right] + \mathbb{P} \left[\|u(t - t_0, \cdot)\|_{\widehat{\rho}} > \Lambda \right] \\ &=: I_1 + I_2. \end{aligned}$$

By Chebyshev's inequality and (1.15), we see that

$$I_2 \leq \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(t - t_0, \cdot)\|_{\hat{\rho}}^2 \right) \leq \frac{1}{\Lambda^2} \mathcal{G}_{\hat{\rho}}(t - t_0; \mu).$$

Because $\mathcal{G}_{\hat{\rho}}(t; \mu)$ is a continuous function for $t > 0$, and because it is also bounded at infinity thanks to Assumption (ii) (see (1.17)), we have that

$$\mathcal{G}_{\hat{\rho}}(t - t_0; \mu) \leq \sup_{t \geq \tau} \mathcal{G}_{\hat{\rho}}(t - t_0; \mu) = \sup_{t \geq t_0} \mathcal{G}_{\hat{\rho}}(t; \mu) < \infty. \quad (4.14) \quad \boxed{\text{E:GtBdd}}$$

Therefore, we can bound I_2 from above with a constant that does not depend on $t \geq \tau$, namely,

$$I_2 \leq \frac{1}{\Lambda^2} \sup_{t \geq t_0} \mathcal{G}_{\hat{\rho}}(t; \mu) < \infty.$$

As for I_1 , with the choice of α and q in (4.12), one can apply Chebyshev's inequality and part (3) of Lemma 4.2 to see that

$$I_1 \leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \mathbb{E} \int_0^{t_0} \|Y_v(s, \cdot)\|_{\hat{\rho}}^q ds \leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \Theta \int_0^{t_0} (\mathcal{G}_{\hat{\rho}}(s + t - t_0; \mu) \mathcal{H}_{\alpha}(s))^{q/2} ds.$$

where the constant Θ does not depend on t . Noticing that Assumption (iii) (see (1.18)) implies that $\mathcal{H}_{\alpha}(s)$ is a continuous function that vanishes at $s = 0$ so that $\int_0^{t_0} \mathcal{H}_{\alpha}^{q/2}(s) ds < \infty$, together with (4.14), we derive the following upper bound for I_2 that is uniform in $t \geq \tau$.

$$I_1 \leq \frac{\sin^q(\alpha\pi)\Theta}{\pi^q \Lambda^q} \left(\sup_{t \geq t_0} \mathcal{G}_{\hat{\rho}}(t; \mu) \right)^{q/2} \int_0^{t_0} \mathcal{H}_{\alpha}^{q/2}(s) ds < \infty.$$

Combining these two upper bounds, we see that

$$\mathbb{P}[v(t_0, \cdot) \notin \mathcal{K}(\Lambda)] \leq \frac{\sin^q(\alpha\pi)\Theta}{\pi^q \Lambda^q} \left(\sup_{t \geq t_0} \mathcal{G}_{\hat{\rho}}(t; \mu) \right)^{q/2} \int_0^{t_0} \mathcal{H}_{\alpha}^{q/2}(s) ds + \frac{1}{\Lambda^2} \sup_{t \geq t_0} \mathcal{G}_{\hat{\rho}}(t; \mu) < \infty,$$

with the upper bound holding uniformly for all $t \geq \tau$. Hence, for any $\epsilon > 0$, by choosing $\Lambda > 0$ big enough such that

$$\frac{\sin^q(\alpha\pi)\Theta}{\pi^q \Lambda^q} \left(\sup_{t \geq t_0} \mathcal{G}_{\hat{\rho}}(t; \mu) \right)^{q/2} \int_0^{t_0} \mathcal{H}_{\alpha}^{q/2}(s) ds + \frac{1}{\Lambda^2} \sup_{t \geq t_0} \mathcal{G}_{\hat{\rho}}(t; \mu) < \epsilon,$$

we can ensure that

$$\mathbb{P}(u(t, \cdot) \in \mathcal{K}(\Lambda)) = \mathbb{P}(v(t_0, \cdot) \in \mathcal{K}(\Lambda)) \geq 1 - \epsilon, \quad \text{for all } t \geq \tau,$$

which proves part (a) of Theorem 1.3. □

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